

8 (a) since momentum before combining is zero
momenta must be equal and opposite after
equal momenta so photon energies equal

B1
B1
B1 [3]

(b) $E = mc^2$
 $= 9.1 \times 10^{-31} \times (3.0 \times 10^8)^2$
 $= 8.19 \times 10^{-14} \text{ (J)}$
 $= (8.19 \times 10^{-14}) / (1.6 \times 10^{-13})$
 $= 0.51 \text{ MeV}$

C1
C1
A1 [3]

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Section B

0 (a) blocks labelled sensing device / sensor / transducer

B1